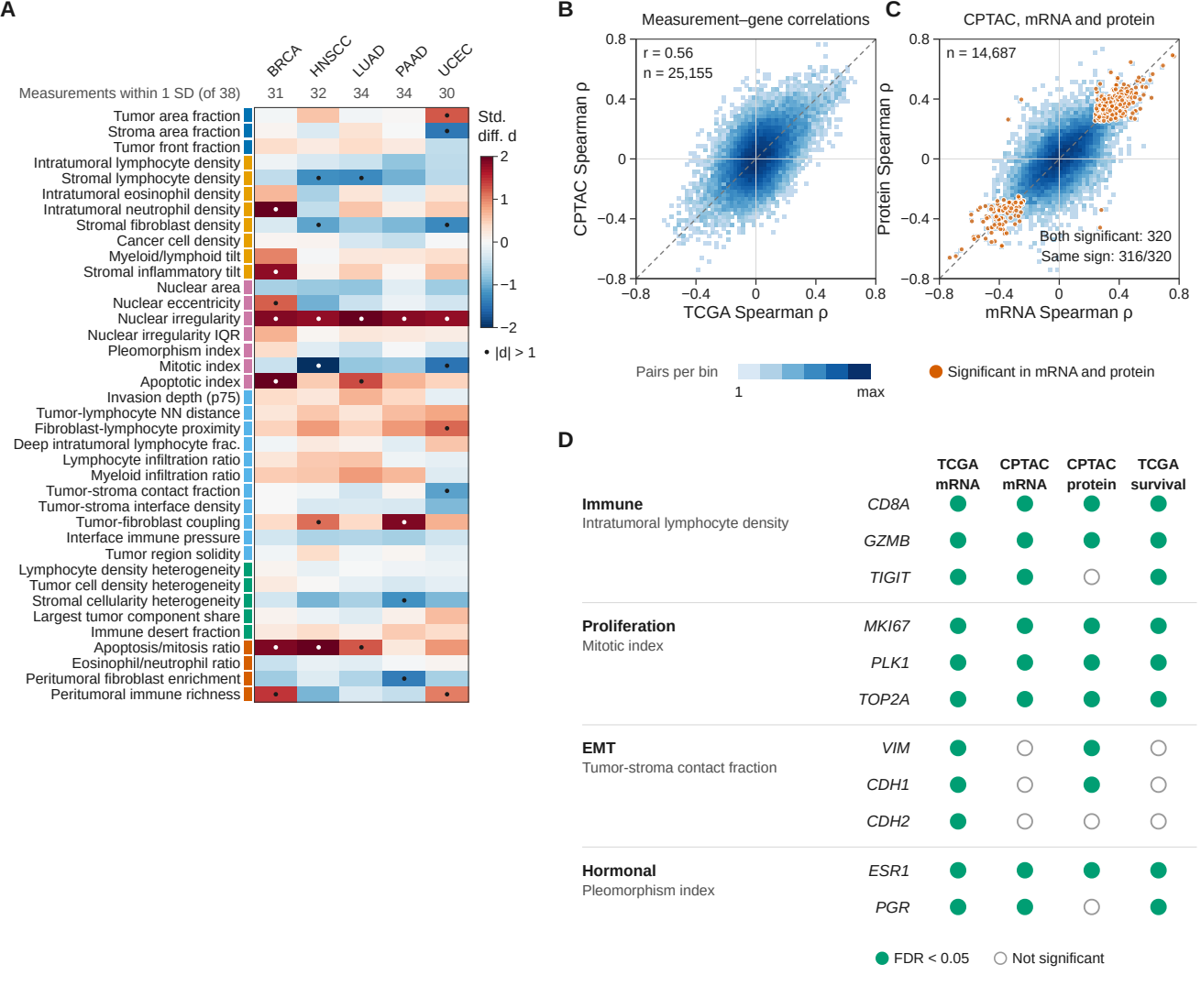

A
BRCA
HNSCC
LUAD
PAAD
UCEC
Measurements within 1 SD (of 38)
31
32
34
34
30
Tumor area fraction
Stroma area fraction
Tumor front fraction
Intratumoral lymphocyte density
Stromal lymphocyte density
Intratumoral eosinophil density
Intratumoral neutrophil density
Stromal fibroblast density
Cancer cell density
Myeloid/lymphoid tilt
Stromal inflammatory tilt
Nuclear area
Nuclear eccentricity
Nuclear irregularity
Nuclear irregularity IQR
Pleomorphism index
Mitotic index
Apoptotic index
Invasion depth (p75)
Tumor-lymphocyte NN distance
Fibroblast-lymphocyte proximity
Deep intratumoral lymphocyte frac.
Lymphocyte infiltration ratio
Myeloid infiltration ratio
Tumor-stroma contact fraction
Tumor-stroma interface density
Tumor-fibroblast coupling
Interface immune pressure
Tumor region solidity
Lymphocyte density heterogeneity
Tumor cell density heterogeneity
Stromal cellularity heterogeneity
Largest tumor component share
Immune desert fraction
Apoptosis/mitosis ratio
Eosinophil/neutrophil ratio
Peritumoral fibroblast enrichment
Peritumoral immune richness
Std. diff. d
2
1
0
−1
−2
• |d| > 1
B
Measurement–gene correlations
r = 0.56
n = 25,155
CPTAC Spearman ρ
TCGA Spearman ρ
0.8
0.4
0
−0.4
−0.8
C
CPTAC, mRNA and protein
n = 14,687
Protein Spearman ρ
mRNA Spearman ρ
Both significant: 320
Same sign: 316/320
Pairs per bin
1
max
Significant in mRNA and protein
D
TCGA mRNA
CPTAC mRNA
CPTAC protein
TCGA survival
Immune
Intratumoral lymphocyte density
CD8A
GZMB
TIGIT
Proliferation
Mitotic index
MKI67
PLK1
TOP2A
EMT
Tumor-stroma contact fraction
VIM
CDH1
CDH2
Hormonal
Pleomorphism index
ESR1
PGR
FDR < 0.05
Not significant